

Learning Per-Layer Fusion for Multi-Control Diffusion Models

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Figure 1: Targeted qualitative case example - a white shirt on a white background

Abstract—Controllable text-to-image diffusion models make it possible to guide image generation with structured signals such as edge maps and depth maps, but practical multi-control pipelines still usually rely on manually chosen global weights for each control branch. This report studies a learned alternative for combining two ControlNet conditions, Canny edges and monocular depth, inside a Stable Diffusion v1.5 generation pipeline. Building on the latent diffusion framework and the ControlNet architecture, the proposed method keeps the Stable Diffusion U-Net and both ControlNet branches frozen and learns a context-conditioned fusion network that predicts residual mixing weights at every ControlNet injection point. The context vector is formed from pooled Canny ControlNet features, pooled Depth ControlNet features, a timestep embedding, and a summarized text embedding, allowing the fusion policy to vary across prompts, images, and diffusion timesteps rather than remaining static. The final implementation uses Stable Diffusion v1.5, ControlNet 1.1 for both Canny and Depth, a Stability AI VAE, and Depth Anything V2 for depth extraction. Training uses the standard diffusion denoising objective on a 20k-image dataset formed by combining 10k samples from DiffusionDB with 10k samples from Flickr30k. Results show that learned fusion is particularly helpful in visually ambiguous cases such as shadows, mirrors, glass,

and weak-edge regions, while remaining comparable to strong baselines in more ordinary cases. The project also reveals a new compositional application in which depth from one image and edge structure from another can be fused to synthesize a new image, though this currently remains limited by the absence of explicit spatial alignment.

I. Introduction and Motivation.

Latent diffusion models have become a standard foundation for high-quality image synthesis because they perform the denoising process in the latent space of an autoencoder rather than directly in pixel space, substantially reducing computational cost while preserving visual fidelity. Stable Diffusion is one of the most widely used practical realizations of this idea, and ControlNet extends such backbones by introducing spatially conditioned residual guidance from signals such as edges, depth, segmentation, and pose.

Among these controls, Canny edges and depth maps are especially attractive because they encode complementary aspects of an image. Canny provides local boundary and contour information, while depth provides coarse geometric layout and relative spatial arrangement. In practice, however, using both controls at once introduces a new problem: how strongly should each control influence generation, and should that balance

remain constant across all layers of the denoising U-Net? Existing practical workflows often answer this question with manually tuned condition scales, which are global, heuristic, and usually selected by trial and error rather than learned from data. The proposal for this project was originally motivated by exactly this limitation and asked whether a learned per-injection fusion mechanism could outperform fixed manual weighting.

As the project evolved, this question became more specific. Early versions used a small fusion MLP that predicted one weight pair per ControlNet injection point. The final version instead uses a context-conditioned fusion module whose output depends on pooled Canny ControlNet features, pooled Depth ControlNet features, timestep embedding, and a summarized text embedding. This means the fusion policy is no longer static even for the same layer: it can change across prompts, input images, and timesteps. Inference is also deterministic under a fixed manual seed, making direct baseline comparisons possible under matched settings.

The central idea of this report is therefore straightforward: instead of manually choosing how much Canny and depth should matter, learn that balance from the denoising objective itself while keeping the expensive pretrained backbones frozen. This framing keeps the method parameter-efficient, makes it practical to train on modest resources, and isolates fusion as the main variable of interest. The final study evaluates the learned fusion network against canny-only, depth-only, and fixed-fusion baselines, and also documents a new compositional use case where depth from one image and edges from another can be combined into a single synthesized output.

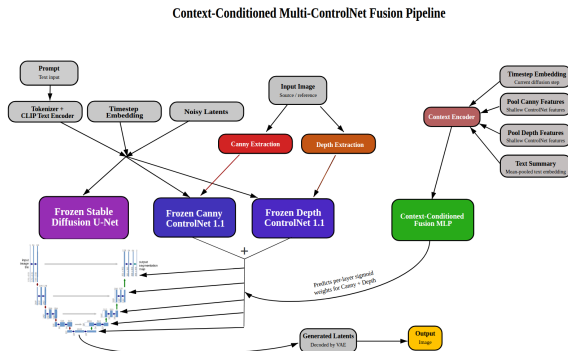
Figure 2: overview of the full pipeline: prompt and image preprocessing, Canny and depth extraction, two frozen ControlNet branches, context encoder, context-conditioned fusion MLP, fused residual injection into frozen Stable Diffusion U-Net.

II. Related Work

The first foundation for this project is the latent diffusion model. Rombach et al. introduced latent diffusion as a way to move diffusion from pixel space into the latent space of a pretrained autoencoder, making high-resolution synthesis substantially more efficient while preserving strong generative quality. Their formulation also established the now-standard text-conditioning pathway through cross-attention, which later systems such as Stable Diffusion inherited directly.

The second foundation is ControlNet. Zhang, Rao, and Agrawala proposed ControlNet as a way to add spatial conditioning to large pretrained text-to-image diffusion models without destroying the backbone’s existing capabilities. Their design locks the pretrained backbone and adds condition-specific trainable pathways connected through zero convolutions, enabling robust control from conditions such as edges, depth maps, segmentation masks, and human pose. Importantly for this project, the original ControlNet paper already demonstrated that multiple conditions can be used, but did not focus on learning a dedicated fusion module for balancing different control branches at every injection point.

In practice, multi-ControlNet usage is heavily shaped by open-source tooling rather than only by papers. The widely used `sd-webui-controlnet` extension popularized on-the-fly use of ControlNet and related injection-based controls in Stable Diffusion interfaces, making it common for users to combine several controls at once. However, these workflows generally rely on user-selected condition strengths rather than an explicitly learned, per-layer, context-dependent fusion policy. In that sense, they highlight the practical importance of the problem addressed here.



Since the original ControlNet, several follow-up works have pushed controllable diffusion in different directions. ControlNet++ improves conditional fidelity by adding a consistency-based reward objective and reports improved controllability for different conditions. Ctrl-Adapter focuses on efficient adaptation of pretrained controls to new image and video diffusion backbones and supports fine-grained multi-condition control. ControlNet-XS revisits the ControlNet communication pathway from a control-systems perspective and proposes a smaller and faster variant. More recently, DC-ControlNet targets richer multi-condition generation by decoupling inter-element and intra-element controls. Together, these works show that controllable diffusion remains active and that multi-condition interaction is still an open design problem.

Relative to this literature, the contribution of the present project is modest but distinct. Rather than training a new control backbone or redesigning the full conditioning architecture, it focuses narrowly on the fusion problem for two specific controls, Canny and depth, under frozen Stable Diffusion and ControlNet backbones. To the best of the sources reviewed here, this exact formulation of context-conditioned per-injection fusion for the Canny-plus-depth pair is not the standard approach in prior ControlNet work. The project therefore fits best as a parameter-efficient study of adaptive multi-control fusion rather than as a replacement for broader control architectures

III. Methods

The final system is built on Stable Diffusion v1.5 as the generative backbone, with two frozen ControlNet 1.1 branches, one for Canny edges and one for depth, and a pretrained VAE for latent encoding and decoding. ControlNet 1.1 is an updated practical release of the original ControlNet family, while Depth Anything V2 is used for monocular depth extraction because it reports stronger fine-detail recovery and robustness than its predecessor - Intel/dpt-large. The underlying Stable Diffusion v1.5 and ControlNets contain over 1.3 billion parameters, satisfying the large-scale model requirement.

For an input image-caption pair, the image is resized and converted into two control representations: a three-channel Canny edge image and a three-channel depth image. In the training pipeline, these controls, together with latents and text embeddings, are precomputed and loaded from tensor files. The code scans a tensor directory for matching `_canny.pt`, `_depth.pt`, `_latent.pt`, and `_text_emb.pt` files, which removes repeated preprocessing from the main training loop.

During training, the Stable Diffusion U-Net and both ControlNets remain frozen, and only the fusion network is optimized. The model samples one timestep and one Gaussian noise tensor per image, adds noise to the latent using the scheduler, runs both ControlNet branches, fuses their residual outputs, and then feeds the fused residuals into the frozen U-Net. The optimization target is the standard denoising MSE between the U-Net noise prediction and the sampled true noise. There is therefore no special handcrafted “fusion loss”; the fusion module is trained indirectly through its effect on denoising quality. This follows the design discussed throughout the project and is visible directly in the training code.

A key implementation detail is how fusion is aligned with the ControlNet architecture. The number of injection points is not chosen manually. Instead, it is determined from the actual ControlNet output structure as the number of down-block residual tensors plus one additional mid-block residual. The fusion module therefore predicts one Canny/depth weight pair for every down-block residual and one final pair for the mid-block residual. This ensures that the learned fusion policy matches the true residual interface of the underlying model.

The main architectural change from the original proposal is that the final fusion model is context-conditioned rather than purely layer-conditioned. The context encoder builds a compact feature vector from four sources: a sinusoidal timestep embedding, a mean-pooled over the 77 CLIP tokens text embedding, globally pooled shallow (first down-block) Canny ControlNet

features, and globally pooled shallow Depth ControlNet features. These four branches are projected and concatenated into a single context vector, which is then provided to the fusion MLP. Each branch is of width 64, producing a 256-dimensional context vector, and the fusion network itself uses learned index embeddings of size 32 followed by a 2-hidden-layer MLP of width 64. In total, the trainable fusion module has only about 130k parameters, keeping the adaptation small relative to the frozen diffusion backbone while still allowing the same layer to receive different Canny/depth balances for different prompts, images, and timesteps.

Another important design choice concerns the output nonlinearity. Earlier experiments used a softmax over the two control weights at each layer. In practice, this caused near-binary routing, with many layers collapsing toward almost pure Canny or pure depth. The final model instead uses independent sigmoid outputs. This removes the forced competition between branches and allows both controls to stay active at the same layer when useful. In the adopted interpretation, values around 1 act as a neutral baseline, values above 1 amplify a branch, and values below 1 suppress it. Project experiments found this formulation to be more stable and more representative of additive fusion than the earlier softmax version.

The final training dataset combines 10,000 images from DiffusionDB with 10,000 images from Flickr30k in order to increase variability in scene layout, object categories, and textual descriptions. Each dataset is split independently using an 80% training and 20% evaluation partition, so both sources contribute to both optimization and held-out testing rather than having one dataset dominate only one stage. This was introduced after the original project proposal, which had focused on DiffusionDB alone. The purpose of the combined dataset is not to build a giant-scale training corpus, but to expose the fusion module to a more diverse range of control situations while keeping the frozen-backbone setup computationally feasible.

IV. Results

The primary baselines considered in this project are canny-only control and depth-only control. The proposed method is compared against these baselines under matched inference settings, and the final implementation uses a manually controlled random seed so that the same prompt and settings produce the same output. This matters because diffusion outputs are otherwise stochastic, making qualitative comparison noisier than necessary. Deterministic inference makes it possible to attribute output differences more confidently to the conditioning method rather than to random sampling.

Training is evaluated with the same denoising MSE used as the objective. In the reported setup, evaluation loss is the average MSE over all evaluation batches. As a result, when evaluation batches contain different sampled noise realizations and timesteps, the epoch-level average can fluctuate even when training is progressing normally. This behavior was observed during the project and is consistent with the stochastic nature of diffusion training. At the same time, when evaluation is performed with a fixed seed and fixed timestep, the loss exhibits a clearer convergence trend, suggesting that the learned fusion module is stabilizing rather than merely oscillating. The project summary and the final code both support this interpretation, including the deterministic evaluation configuration with a fixed seed and fixed timestep.

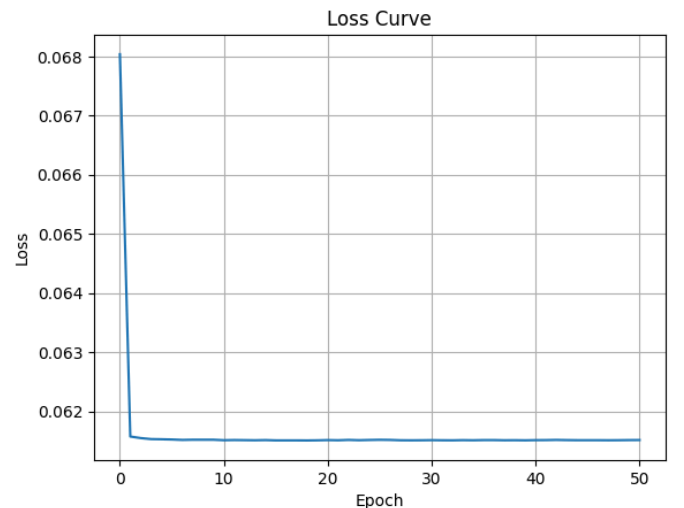


Figure 3: Evaluation loss with a fixed seed and fixed timestep over one batch..

Qualitatively, the strongest benefits of learned fusion were observed in cases where one control signal alone was incomplete. In scenes involving shadows, mirrors, glass, and weak or low-contrast edges in [Fig. 1], canny-only control often missed meaningful structure, while depth-only control tended to be too coarse to preserve important local boundaries. The context-conditioned fusion model performed best in exactly these ambiguous cases because it could preserve local contour information where edges were reliable while still using depth when broader geometric cues were more informative. In ordinary scenes with strong boundaries and clear layout, the learned method was usually comparable rather than dramatically superior, which is still a useful result: it suggests the adaptive fusion does not significantly harm cases already handled well by simpler controls.

The learned weights also support the design decisions made during development. The earlier softmax model routinely collapsed to extreme layer-wise preferences, effectively turning many injection points into hard gates. In contrast, the sigmoid-based model preserved more balanced and interpretable additive behavior. Rather than forcing one branch to win at each layer, it allowed the model to attenuate or amplify both branches as needed. This was especially important after moving to the context-conditioned formulation, since the goal was no longer just to learn a global layer preference but to adapt the balance dynamically across samples and timesteps. In that sense, the final behavior is better understood as conditional routing with soft additive control rather than winner-take-all selection.

A second outcome of the project was the discovery of a new application. By taking the depth map from one image as a background/layout prior and the Canny edges from another image as a foreground/structure prior, the system can synthesize a new image that combines aspects of both sources. This is an interesting form of compositional image generation. Its current limitation is that the two control inputs are not spatially aligned explicitly. Successful examples therefore require the user to choose source images whose object placement and geometry already

approximately match. This limitation suggests an obvious future direction: incorporate explicit spatial awareness or region-aware correspondence so that the control signals can be aligned before fusion.



Figure 4: deterministic comparison grid under fixed seed: input depth and canny, depth-only result, Canny-only result, learned context-conditioned fusion result.

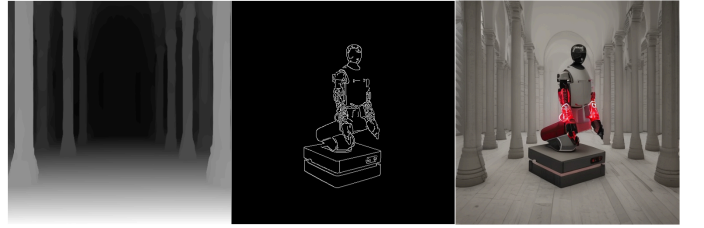


Figure 5: compositional application: source image for depth, source image for Canny, and resulting fused output.

VI. Conclusion

This work investigated adaptive fusion for multi-ControlNet image generation. Starting from the observation that practical multi-control workflows still often depend on manually selected global condition scales, the project proposed a learned fusion module that combines Canny and depth residuals inside a frozen Stable Diffusion v1.5 pipeline. The final version of the system is context-conditioned rather than static: fusion weights depend on pooled Canny and depth ControlNet features, timestep embedding, and text embedding summary, allowing the control balance to vary across prompts, images, and denoising stages. The project also found that sigmoid-based branch outputs are better suited than softmax for this setting because they avoid near-binary collapse and better support additive multi-control fusion.

Empirically, the method is most valuable in visually ambiguous cases where neither control is sufficient on its own, especially shadows,

reflections, transparent surfaces, and weak-edge regions. In more standard cases, performance remains similar to strong simpler baselines, which is itself useful because it suggests that adaptive fusion can provide selective gains without broadly sacrificing control quality. Deterministic seeded inference and fixed-timestep evaluation further helped make these comparisons more reliable.

The project also points toward two concrete future directions. First, the current compositional application would benefit from explicit spatial awareness so that controls extracted from different source images can be aligned more reliably. Second, the present work uses only two controls, while practical generation systems increasingly involve larger control dictionaries and more complex condition interaction. Extending context-conditioned fusion from two controls to many controls would move the method closer to general multi-condition routing, which remains an active area of controllable diffusion research.

VII. References

- [1] Rombach, R., Blattmann, A., Lorenz, D., Esser, P., & Ommer, B. High-Resolution Image Synthesis with Latent Diffusion Models. [Online]. Available: <https://arxiv.org/abs/2112.10752>.
- [2] Zhang, L., Rao, A., & Agrawala, M. Adding Conditional Control to Text-to-Image Diffusion Models. [Online]. Available: <https://arxiv.org/abs/2302.05543>.
- [3] Mikubill, “Mikubill/SD-webui-controlnet: Webui extension for ControlNet,” GitHub. [Online]. Available: <https://github.com/Mikubill/sd-webui-controlnet>.
- [4] Depth Anything Team. Depth Anything V2. [Online]. Available: <https://arxiv.org/abs/2406.09414>
- [5] Li, M. et al. ControlNet++: Improving Conditional Controls with Efficient Consistency Feedback. [Online]. Available: <https://arxiv.org/abs/2404.07987>
- [6] Lin, H. et al. Ctrl-Adapter: An Efficient and Versatile Framework for Adapting Diverse Controls to Any Diffusion Model. [Online]. Available: <https://arxiv.org/abs/2404.09967>
- [7] Zavadski, D., Feiden, J.-F., & Rother, C. ControlNet-XS: Rethinking the Control of Text-to-Image Diffusion Models as Feedback-Control Systems. [Online]. Available: <https://arxiv.org/abs/2312.06573>
- [8] Yang, H. et al. DC-ControlNet: Decoupling Inter- and Intra-Element Conditions in Image Generation with Diffusion Models. [Online]. Available: <https://arxiv.org/abs/2502.14779>